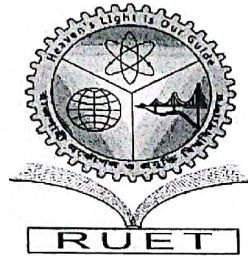


Heaven's Light is Our Guide

Rajshahi University of Engineering and Technology



Department of Electrical & Computer Engineering

Electrical Machine-I Sessional

ECE 2208

(0.75 Credit)

INDEX

SL.	Name of Experiments
01	Open circuit test and short circuit test of a single-phase transformer.
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05	Study of a 1- ϕ induction motor.
06	Starting an Induction Motor by applying star-delta starter.

LAB INFORMATION

In the laboratory we use:

01. DC Generator.
02. DC Motor.
03. Induction Motor
04. Synchronous machine
05. Starter.
06. Techo Meter.
07. Resistors.
08. Voltmeter.
08. Ammeter.
10. Wattmeter
11. Variac.

Study of the rating of different machines:

DC Generator: The machine, which can transform mechanical energy to electrical energy, is called generator. If the output is DC voltage, then that machine is called dc generator.

DC motor: The machine which can transform electrical energy to machine energy is called motor. If the input is DC, then this motor is called DC motor.

Rating Table:

DIRECT CURRENT MACHINE (DC GENERATOR / MOTOR)

KW	2 kW	Volts- 250/250 V	AMP – 8.0 A
RPM	1450	WOUND COMPOUND	
FLD-AMPS	0.467 A	AS SH GEN	FLD-OHMS 25 C 350 Ω
DUTY CONT	60 C RISE	ENCL DP	SERVICE FACT 1.5
ARMATURE RESISTANCE - 1.68 Ω			

DIRECT CURRENT MACHINE (DC GENERATOR / MOTOR)

KW	4 1/2 kW	Volts- 250/250 V	AMP – 18.0 A
RPM	1450	WOUND COMPOUND	
FLD-AMPS	1.0 A	AS SH GEN	FLD-OHMS 25 C 152.8 Ω
DUTY CONT	60 C RISE	ENCL DP	SERVICE FACT 1.15

SUIT	AS	5 HP	1500/ 3000 RPM 240 V
ARMATURE (dc) RESISTANCE - 1.72 Ω			

SYNCHRONOUS MACHINE (CONSTANT SPEED MACHINE)

TYPE	SJ		FRME 254Z
1500 RPM		50 CYCLE	50 C RISE
GENERATOR 4KVA 0.8 P.F		220/240 V 3 PHASE	19.2 /9.6 A FL AMP
MOTOR 6 HP 1.0 P.F		120 /240 V 3 HP	22 / 11 A FL AMP
FILD-AMP MAX 2.1			FILD VOLTS 125 V
Line to line stator terminal (dc)Resistance = 0.75 Ω			

Induction Motor: An AC motor, whose speed is varied with load.

3 – PHASE INDUCTION MOTOR

HP 3	SERVICE FACTOR 1.15
FL RPM 1450	
VOLTS 220 /440	PHASE 3
FL AMPS 9.1 /4.55 A	CYCLE 50
TYPE M	FRAME 254 U
C RISE 40	TIME RATING CONT
SEC VOLTS 145	SEC AMP 10
Line to line stator terminal (dc)Resistance = 0.75 Ω	

SINGLE PHASE INDUCTION MOTOR :

TYPE	DE 1028
RATED VOLTAGE	220 V
RATED CURRENT	6.6 A
RATED POWER	1 kW
RATED SPEED	2820 RPM
FREQUENCY	50 HZ
INSULATION	CLASS E
P.F.	COS ϕ = 0.90
Main Winding (dc) Resistance =	3.95 Ω
Auxiliary Winding (dc) Resistance =	9.32 Ω

Experiment No: 01

Name of the Experiment: Open circuit test and short circuit test of a single-phase transformer.

Theory: We know that there are two types of losses in a transformer, such as copper loss and core loss. Copper loss is a variable loss and depends on current and core loss is a fixed loss and depends upon the voltage. Now the efficiency of a transformer is defined as the ratio of output power to input power. Again the input power is equal to the sum of output power and losses. The losses can be found out with open circuit test and short circuit test. The copper loss can be obtained by short circuit test and core loss can be obtained by the open circuit test. For Short Circuit Test first 0 voltage is applied to high voltage side and then the voltage is increased until for which the rated current flows in low voltage side. Now the data or reading of the meters are taken. Again for Open Circuit Test 0 voltage is applied to low voltage side and then the voltage is increased until for which the rated voltage is obtained from high voltage side. Now the data or reading of the meters are taken.

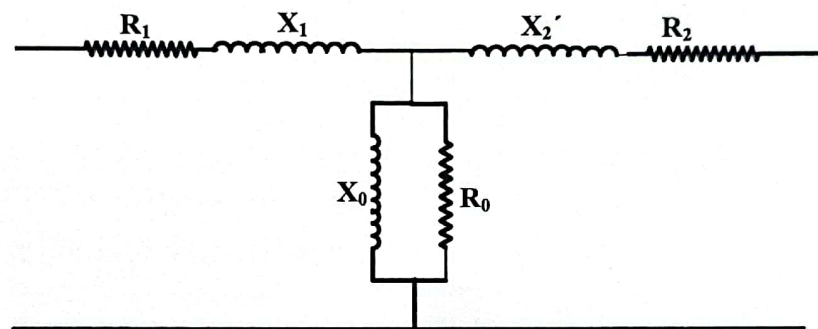
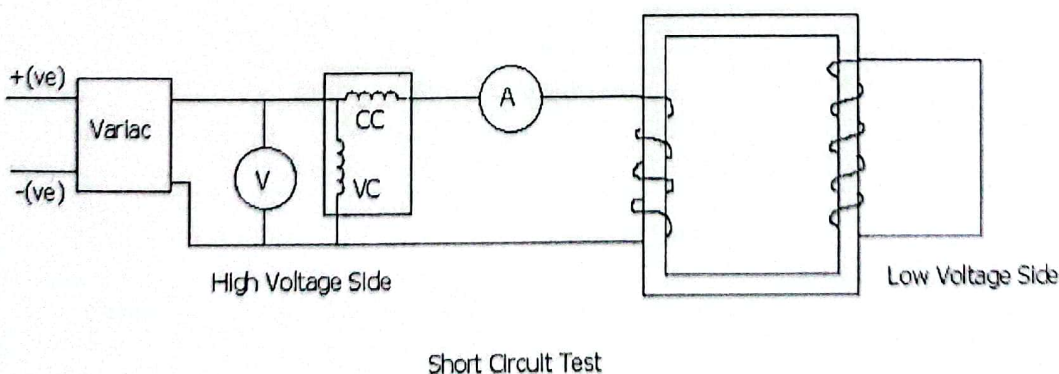


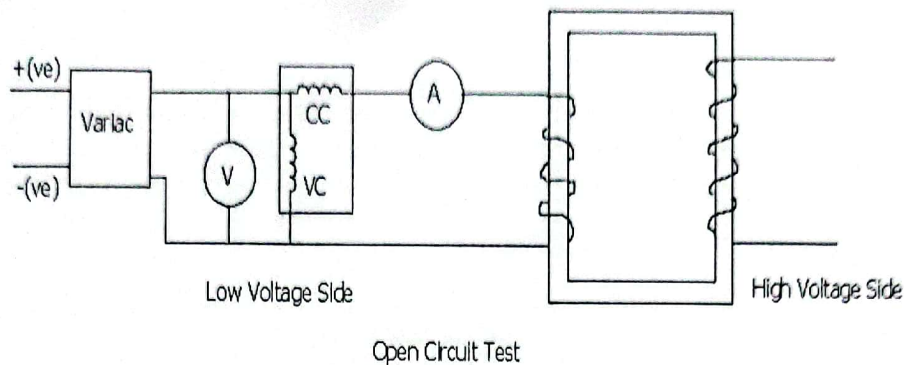
Fig. Equivalent circuit of Transformer

Apparatus:

01. Transformer (150 V per coil, 1A)
01. Voltmeter (0-450V).
02. Ammeter (0-2A).
03. Wattmeter (0-1200W).
04. Variac (0-240 V)

Circuit Diagram:





Data Table:

Test	Voltage	Current	Power
Short circuit test	7.56	1.00	3.5
Open circuit test	100	0.21	16.5

Calculation:

For short circuit test the applied voltage is very low and core loss is negligible. So it is assumed that current through R_o & X_o is zero

$$\text{So we can write, } R_{SCT} = R_1 + R_2' = P / I_{SCT}^2 =$$

$$Z_{SCT} = V_{SCT} / I_{SCT} =$$

$$X_{SCT} = X_1 + X_2' = \sqrt{(Z_{SCT}^2 - R_{SCT}^2)} =$$

For Open circuit test Cu Loss is negligible i.e. loss in R_1 is negligible and voltage drop at R_1 & X_1 is negligible. In this case Loss is assumed only in the core circuit

$$R_{OCT} = R_o = \frac{V_{OCT}^2}{P} =$$

$$G_{OCT} = \frac{1}{R_{OCT}} =$$

$$Z_{OCT} = Z_o = \frac{V_{OCT}}{I_{OCT}} =$$

$$Y_o = \frac{1}{Z_o} =$$

$$B_o = \sqrt{(Y_o^2 - G_o^2)} =$$

$$X_o = \frac{1}{B_o} =$$

$$\% \text{ Efficiency} = (\text{Output} / \text{input}) \times 100\%$$

$$\% \text{ Efficiency} = (\text{Input} - \text{Losses}) \times 100 / \text{Input} \% = (200 - 20) \times 100 / 200 = 90\%$$

$$\text{Efficiency} = \text{Output} / (\text{Output} + \text{losses})$$

Discussion:

Experiment No: 02

Name of the Experiment: Parallel operation of transformers.

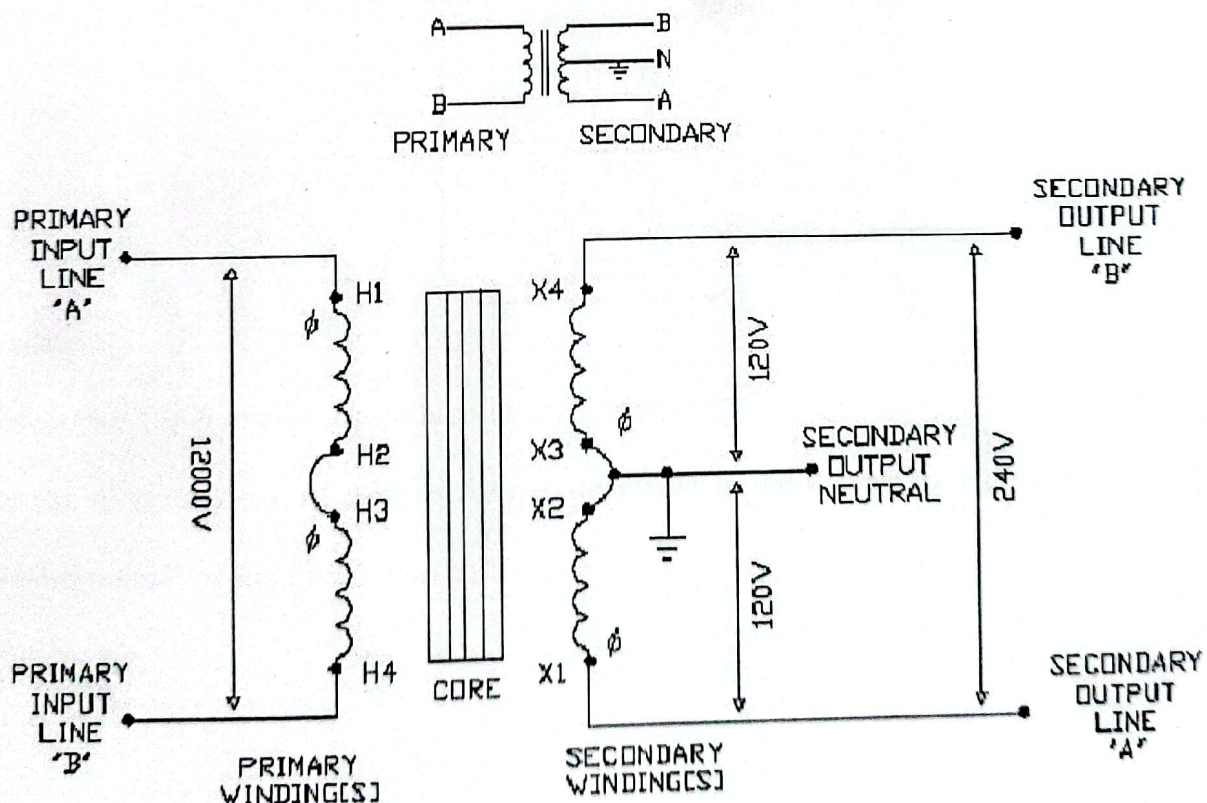
Theory:

When in a power system the demand for load increases, a single transformer may not be able to supply the extra load. In that case, another transformer is connected in parallel with the existing transformer to share the load. In order to make two transformers parallel, some conditions have to be fulfilled. These are:

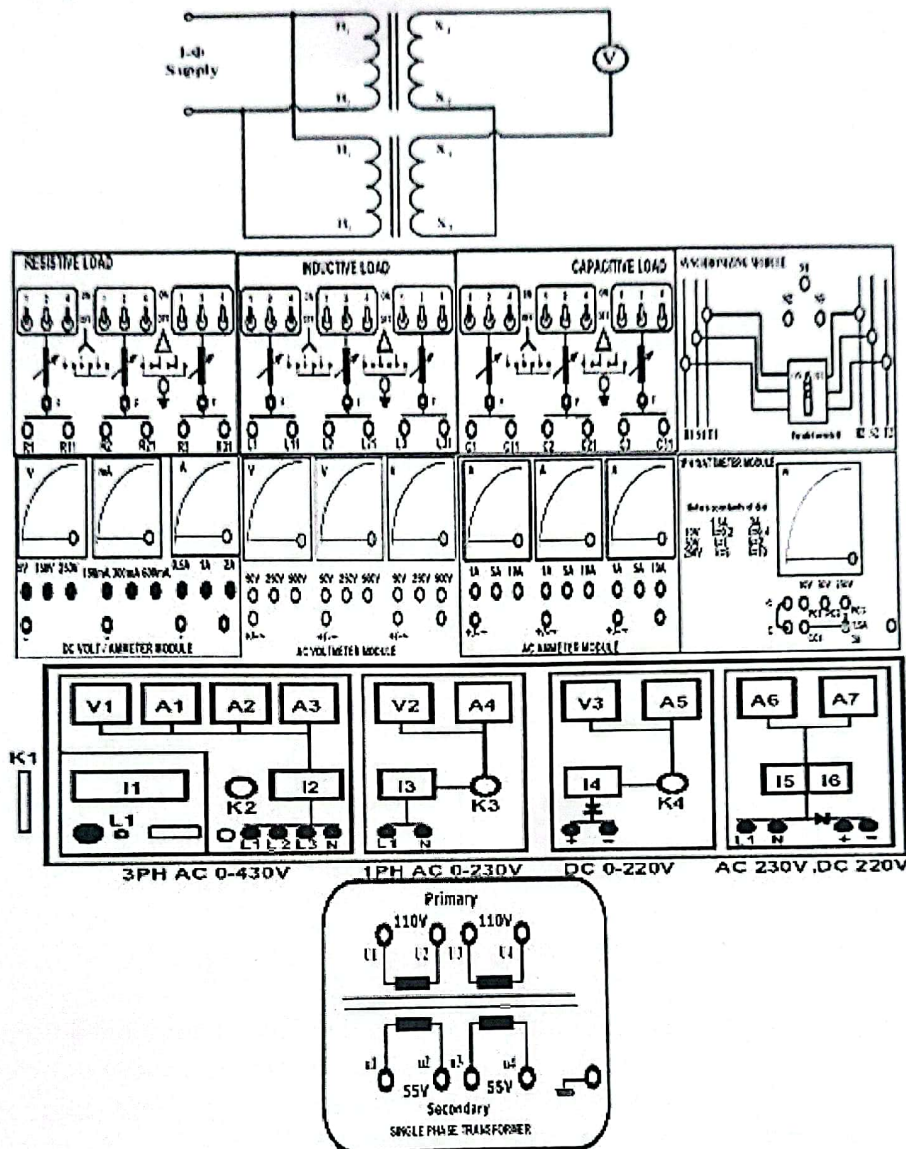
- 1) Terminal voltages on the primary and secondary side should be identical.
- 2) The relative polarities on the primary and secondary sides should be identical.
- 3) Preferably R/X ratio of both the transformers should be same.
- 4) Primary windings of the transformer should be suitable for the supply system voltage and frequency.

Apparatus:

1. 1PH Transformer 2 pieces
2. AC Ammeter Module 0-1A
3. AC Voltmeter Module 0-250 V
4. DC ammeter (0 to 600 mA)
5. Load 100 watts - 2 pieces.
6. Clamp-on-meter -1 piece
7. DC Supply – 9 V
8. Connecting Cables

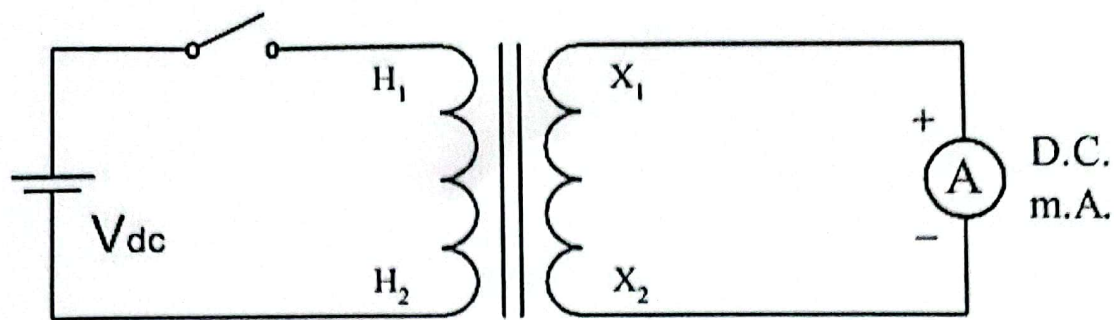


Connection Diagram:



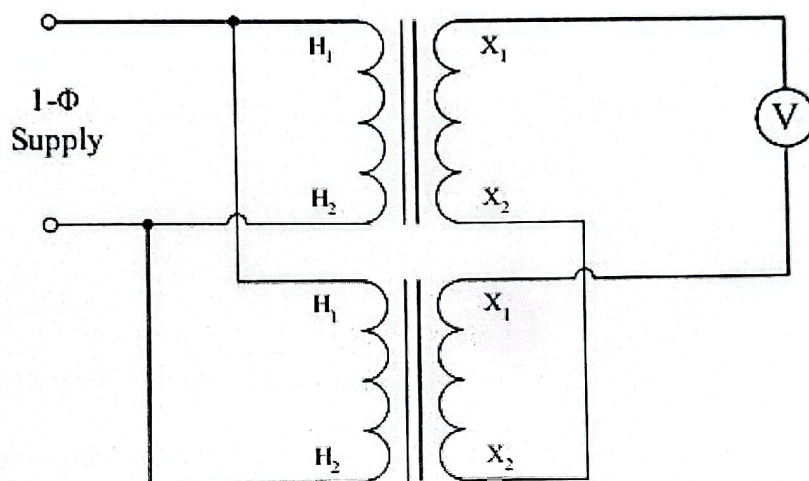
Procedure:

1. Select two 1-φ transformers of identical manufacturer.
2. On the secondary side of the transformer, determine R and X with an R-L-C meter.
3. Determination of Relative Polarity:

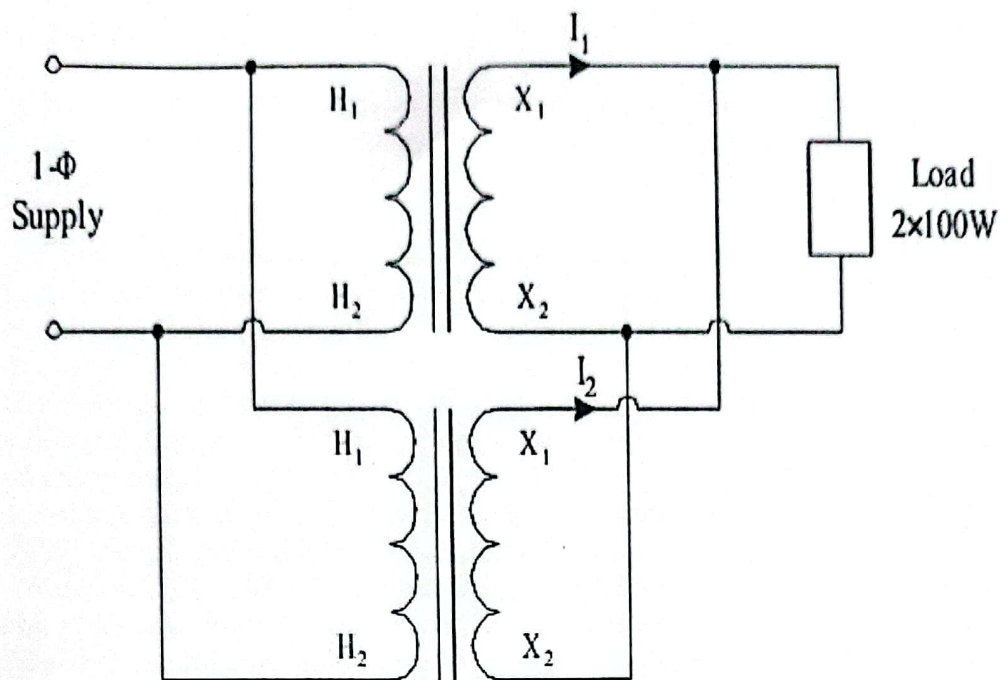


Connect a D.C. mA with polarity as shown on the secondary side, and a battery through a switch on the primary side. Push the switch (give a kick). If the mA deflects in the positive direction, then H_1 and X_1 have the same mode of winding. Repeat the step for TR2 and confirm that the modes of windings are identical.

4. Connection for Parallel Operation:



- Connect a voltmeter and connect the transformers as shown. If the polarities are correct, the voltmeter should read zero.
 - Disconnect the supply, remove the voltmeter and connect the two secondary terminals (X_1 , X_1) of TR1 and TR2. With a clamp-on-meter, check if there is any circulating current. If there is any, note this circulating current.
5. Now, re-connect TR1 and TR2 as in step 4(a). On the secondary side, connect 2 load of 100W each.



With clamp-on-meter, determine the currents supplied by TR1 and TR2

For TR1: $R_1 = X_1 =$

For TR2: $R_2 = X_2 =$

Voltmeter reading on step 4:

clamp-on-meter reading on step 5:

Currents supplied by TR1 =

Currents supplied by TR2 =

1. Discuss on the value of circulating current found in step 4(c).
2. Why is parallel operation necessary? Why is relative polarity test necessary for parallel operation?
3. Why the voltmeter gives zero reading if the polarities are same at step 4(a)?

Experiment No: 03

Name of Experiment: Construction of three-phase transformer using three single-phase transformer and observation of line voltage & phase voltage and line current & phase current relation in primary and secondary winding with balance and unbalance loading.

Theory: A three-phase transformer of both Δ and Y configuration can easily be constructed by using single phase transformer. For connecting the transformers of single phase, we need to know the polarity of each coil. After knowing the polarity of each coil 3- phase connection can be made properly. There are many transformers still in service whose terminal does not have the standard markings. In order to interconnect these transformers with others, either for single-phase parallel operation or for polyphase operation, it is necessary to determine the relative polarities of the terminals.

Y- Y connection without primary neutral grounded works satisfactorily only if the load is balanced. With the unbalanced load to the neutral, the neutral point shifts thereby making the three line-to-neutral (i.e. phase) voltages unequal. The effect of unbalanced loads can be illustrated by placing a single load between phase (or coil) a and the neutral on the secondary side. The power to the load has to be supplied by primary phase (or coil) A. This primary coil A cannot supply the required power because it is in series with primaries B and C whose secondaries are open. Under these conditions, the primary coils B and C act as very high impedances so that primary coil can obtain but very little current through them from the line. Hence, secondary coil a cannot supply any appreciable power. In fact, a very low resistance approaching a short-circuit may be connected between point a and the neutral and only a very small amount of current will flow. This, as said above, is due to the reduction of voltage E_{an} because of neutral shift. In other words, under short-circuit conditions, the neutral is pulled too much towards coil a . This reduces E_{an} but increases E_{bn} and E_{cn} (however line voltage E_{AB} E_{BC} and E_{CA} are unaffected). On the primary side, E_{AN} will be practically reduced to zero whereas E_{BN} and E_{CN} will rise to nearly full primary line voltage. This difficulty of shifting (or floating) neutral can be obviated by connecting the primary neutral (shown dotted in the figure) back to the generator so that primary coil A can take its required power from between its line and the neutral.

Apparatus:

01. Transformer -3 (1- ϕ , 150V, 1A each coil)
02. 3- phase Variac. (0-50V)
03. Voltmeter. (0-450V)
04. Ammeter. (0-2A)
05. Variable Resistor(107 Ω , 2A)

Circuit Diagram:

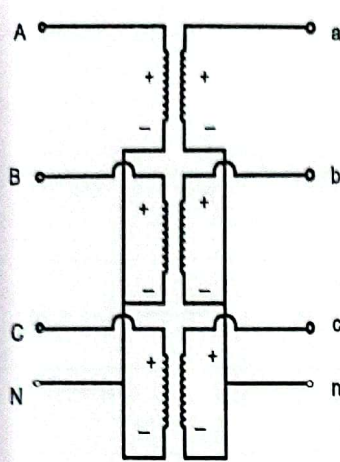


Fig1: Y-Y Connection,

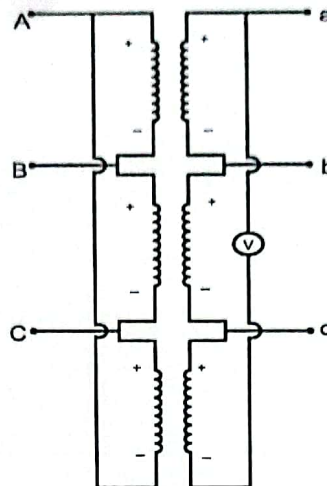


Fig2 polarity test for Δ - Δ connection

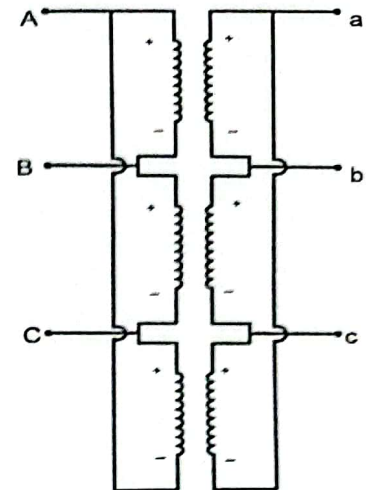


Fig2 : Δ - Δ connection

Fig. Construction of three phase transformer using single phase transformer:

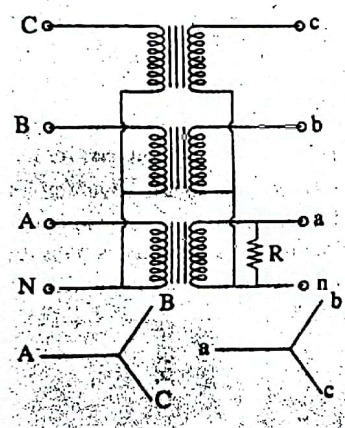


Fig1: Unbalanced load in Y-Y connection without primary neutral grounded.

Precautions:

01. Care should be taken so that current in the coils does not exceed the rated value.
02. Winding should be given the rated or less than the rated supply.
03. Polarity of each coil should always be tested before construction of 3- ϕ transformer.

Data Table:

For Y-Y connection

Sl. NO.	V_{AN}	V_{BN}	V_{CN}	V_{AB}	V_{BC}	V_{CA}	V_{an}	V_{bn}	V_{cn}	V_{ab}	V_{bc}	V_{ca}
01	158	170	164	290	290	290	84	88	86	144	144	144
02	134	132	134	229	230	232	67	66	67	114	114	115

For Δ - Δ connection

Sl. NO.	V_{AB}	V_{BC}	V_{CA}	V_{ab}	V_{bc}	V_{ca}	I_a	I_b	I_c	I_{an}	I_{bn}	I_{cn}
01	210	215	214	106	108	108	0.97	1.0	0.94	0.55	0.58	0.57

For balance /unbalance loading, in Y-Y connection with / without primary neutral grounded

Condition s	V_A N	V_B N	V_C N	V_A B	V_B C	V_C A	V_a n	V_b n	V_{cn}	V_a b	V_{bc}	V_{ca}	I_{cn}
Balanced/ primary neutral grounded	158	170	164	290	290	290	84	88	86	144	144	144	
Unbalance d & primary neutral ungrounde d	250	198	86	290	290	290	128	100	38	144	144	144	
Balanced/ primary neutral grounded	134	132	134	229	230	232	67	66	67	114	114	115	
Unbalance d & primary neutral ungrounde d	207	178	52	230	231	233	103	88	25	114	114	115	

Discussion:

Experiment No: 04

Name of Experiment: Blocked rotor test & no-load test of 3-phase induction motor and determination of circuit parameters.

Theory:

Blocked Rotor Test:

It is also known as locked-rotor or short-circuit test. This test is used to find

1. short-circuit with *normal* voltage applied to stator
2. power factor on short-circuit
3. total leakage reactance X_{01} , of the motor as referred to primary (*i.e.* stator)
4. total resistance of the motor R_{01} as referred to primary.

In this test, the rotor is locked (or allowed very slow rotation) and the rotor windings are short-circuited at slip-rings, if the motor has a wound rotor. Just as in the case of a short-circuit test on a transformer, a reduced voltage (up to 15 or 20 per cent of normal value) is applied to the stator terminals and is so adjusted that full-load current flows in the stator. As in this case $s=1$, the equivalent circuit of the motor is actly like a transformer, having a short-circuited secondary. The values of current, voltage and power input on short circuit are measured by the ammeter, voltmeter and wattmeter connected in the circuits.

W = total power input on short-circuit

V_s = line voltage on short-circuit

I_s = Line current on short-circuit.

Now, the motor input on short-circuit consists of mainly stator and rotor Cu losses. Core-loss, which is small due to the fact that applied voltage is only a small percentage of the normal voltage.

No load test:

In this test rated voltage is applied to the induction motor without any mechanical load and the speed is assumed synchronous speed. In practice, it is not feasible to run the induction motor synchronously for getting R_0 and X_0 . Instead the motor is run without any external mechanical load on it the speed of the rotor would not be synchronous, but very much near to it; so that for all practical purposes, the speed may be assumed synchronous. The no load test is carried out with different values of applied voltage below and above the value of normal voltage. The power input is measured by two watt meters, As motor is running on light load, the p.f. would be low *i.e.* less than 0.5, hence total power input will be the difference of the two wattmeter readings W_1 and W_2 . The readings of the total power input W_0 and total current I_0 and voltage V_0 . The no-load input W_0 to the motor consists of small stator Cu loss $3I_0^2 R_1$, stator core loss and friction and windage. The stator core loss and friction and windage losses *are* collectively known as fixed losses, because they are independent of load.

Apparatus:

- 1) Three phase induction motor (440V, 4.55 A, 1420 rpm).
- 2) Ammeter (0-10 A).
- 3) voltmeter (0-450v).
- 4) Wattmeter (450 V, 10A)
- 5) Ohm meter.

Circuit Diagram:

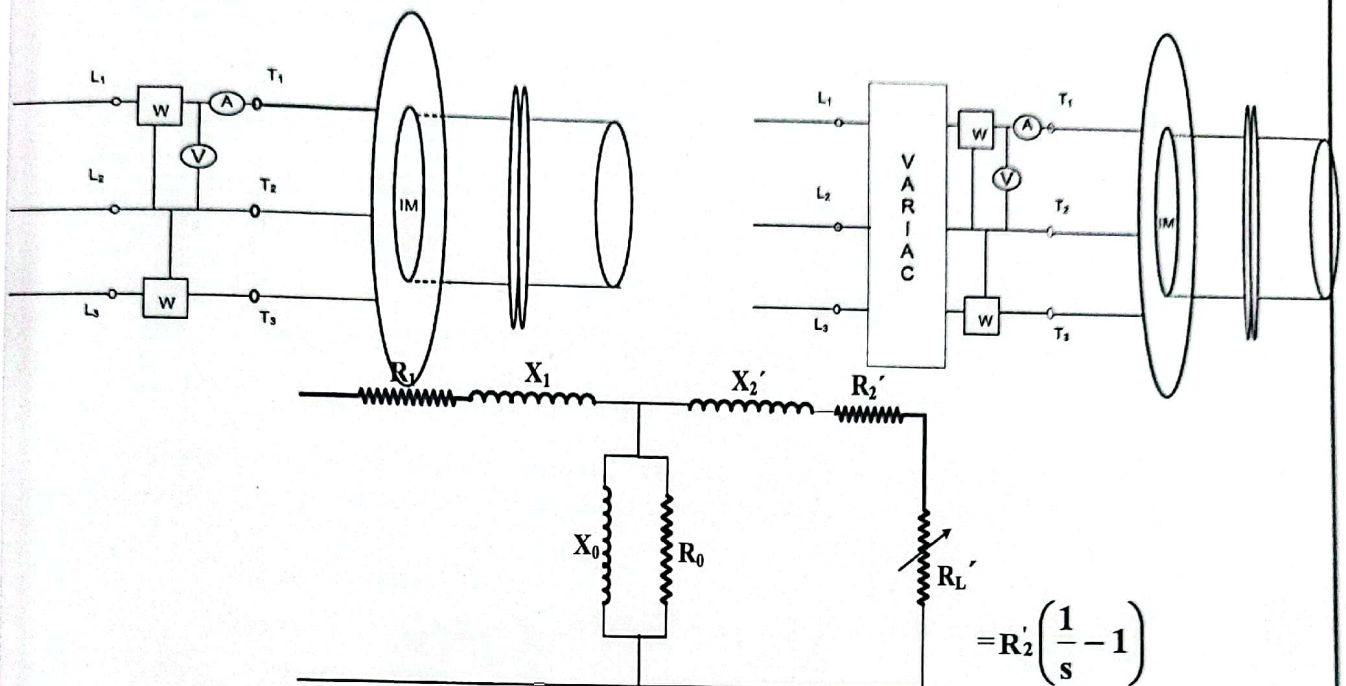


Fig. Electrical equivalent circuit of 3- ϕ induction motor.

Data Table:

For Blocked rotor Test

S.NO.	V_s	I_s	W_1	W_2	W_T	$R_{01} = \frac{P}{3I_s^2}$	$Z_{01} = \frac{V_0}{\sqrt{3} I_0}$	$X_{01} = \sqrt{Z_{01}^2 - R_{01}^2}$
01	99	4.55	255	140	395	6.36	12.56	10.83

No load test

S.NO	V_0	I_0	W_1	W_2	W_T	P_{CORE}	$R_0 = \frac{V_0^2}{P_0}$	$Z_0 = \frac{V_0}{\sqrt{3} I_0}$	$B_0 = \sqrt{Y_0^2 - G_0^2}$	$X_0 = \frac{1}{B_0}$
01	440	2.15	200	820	620	458.4 7	422.27	118.16	8.125×10^{-3}	123.07
02	400	1.95	110	690	580					
03	354	1.80	35	580	545					
04	270	1.6	70	420	490					

DC resistance test of Stator winding : $R_{T-T} = 4.18 \Omega$

Calculation:

For Block rotor test the applied voltage is lower than the rated voltage and core loss is negligible. So it is assumed that current through R_0 & X_0 is zero

So we can write, $R_s = R_1 + R_2' = P/3I_s^2 =$

For Y connected induction motor $R_{1dc} = \frac{R_T}{2} = 2.09$

$$R_1 = 1.4 \times R_{1dc} = 2.926$$

$$\text{So, } R_2' = R_s - R_1 =$$

$$Z_s = V_s / \sqrt{3} I_s =$$

$$X_s = X_1 + X_2' = \sqrt{(Z_s^2 - R_s^2)} =$$

$$\text{It is assumed that } X_1 = X_2' = \frac{X_s}{2}$$

For No Load test there are mainly core loss, very little Cu Loss and friction & windage loss. It is assumed that the voltage drop at R_1 & X_1 is negligible. In this case Core Loss is

$$P_{F+W} = K \times N = 148 \quad (N = 1480, \text{ Assuming, } K = 0.10)$$

$$\text{Stator Cu loss} = I_0^2 \times R_1$$

$$P_{CORE} = W_T - I_0^2 \times R_1 - P_{F+W} = 620 - 2.15^2 \times 2.926 - 148 = 458.47$$

$$R_0 = \frac{3V_{OCT}^2}{3P_{CORE}} = \frac{V_{OCT}^2}{P_{CORE}} =$$

$$G_0 = \frac{1}{R_0} =$$

$$Z_0 = \frac{V_0}{\sqrt{3}I_0} =$$

$$Y_0 = \frac{1}{Z_0} =$$

$$B_0 = \sqrt{(Y_0^2 - G_0^2)} =$$

$$X_0 = \frac{1}{B_0} = 123.07$$

The rated voltage and current of the tested induction motor are 440V and 4.55 A respectively. So if it is assumed that at rated condition p.f is 0.80

$$\text{Then the input power is} = \sqrt{3} V_L I_L \cos \theta = 2774.05 \text{ W}$$

$$\text{And Losses} = P_{CU} + P_{CORE} + P_{F+W} = 1001.5 \text{ W}$$

$$\% \text{ Efficiency} = (\text{Output} / \text{input}) \times 100\%$$

$$\% \text{ Efficiency} = (\text{Input} - \text{Losses}) \times 100 / \text{Input} \% = 63.9 \%$$

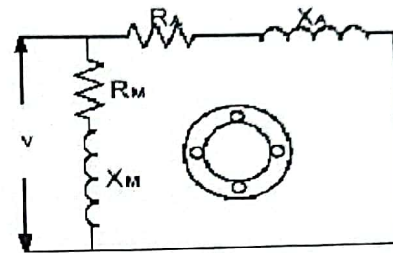
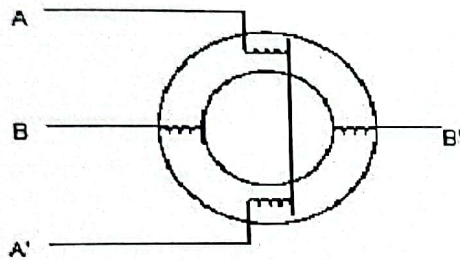
$$\text{Efficiency} = \text{Output} / (\text{Output} + \text{losses}) = 63.9\%$$

Discussion:

Experiment No: 05

Name of Experiment: Study of a 1- ϕ induction motor.

Theory: The single-phase induction motor when fed from 1- ϕ supply, its stator winding produces a flux which is alternating i.e. which alternates along one space axis only. An alternating or pulsating flux acting on a stationary squirrel-cage rotor cannot produce rotation. That is why single-phase induction motor is not self-starting. But if the rotor of such machine is given initial start by hand or somewhat other means in either direction, then immediately torque arises and accelerates to its final speed. To make the motor self starting another winding called a starting/auxiliary winding is placed 90 electrical degrees apart from the main or running winding and connected in parallel across the 1- ϕ supply.



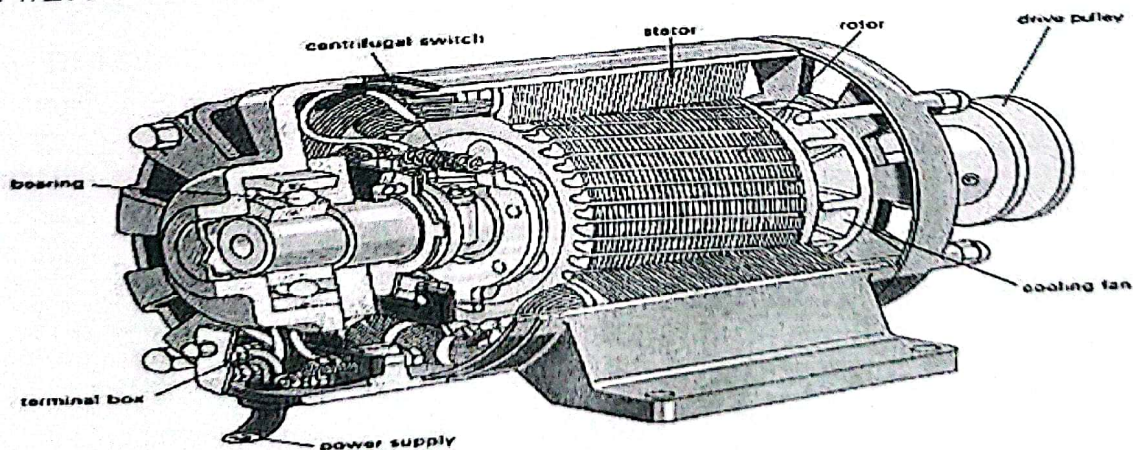
AA/ = Main winding.

BB/ = Auxiliary winding.

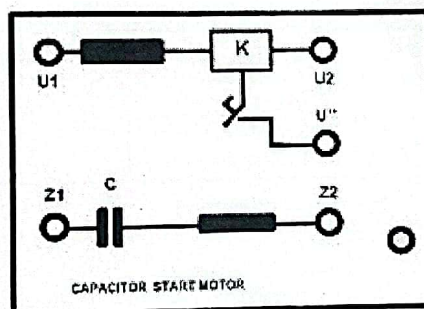
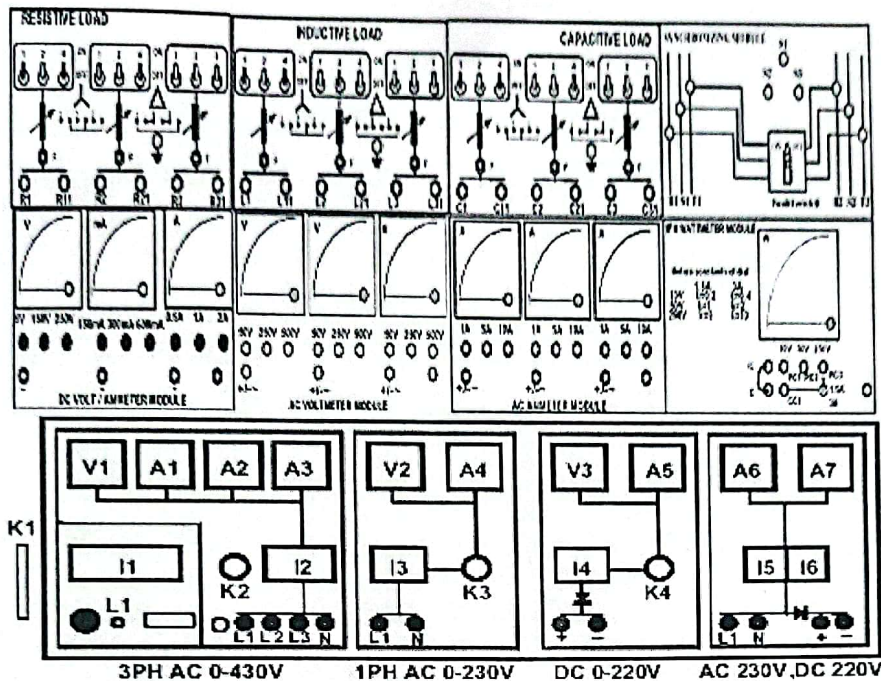
Two theories have been put forward to explain the operation of 1-phase Induction motor. The theories are: 1) Double Revolving Field theory. 2) Cross-field theory.

Apparatus:

1. LC meter
2. 1- ϕ Variac
3. Wattmeter
4. AC Voltmeter (0-300 V)
5. AC Ammeter (0-2.5 A)
6. Tachometer
7. Wire for connection



Connection Diagram:



Procedure:

A) To determine the terminals of running and starting winding:
Normally 3 or 4 terminals are available on the motor terminal box. If one terminal of both starting and running windings are internally connected, then 3-terminals are available on the terminal box. The auxiliary winding is designed to have higher resistance/reactance ratio than the main winding so that the current in the auxiliary winding leads the current in the main winding.

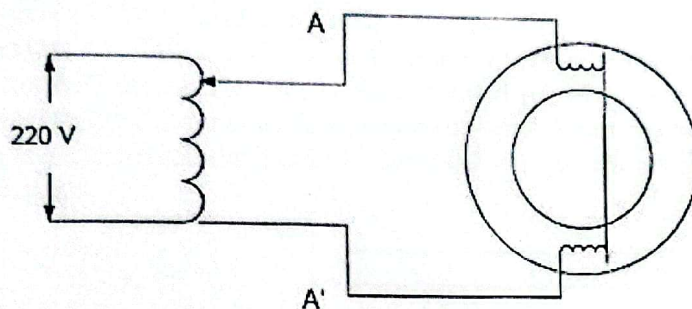
- 1) Take any two-coil terminals and determine its resistance by a sensitive ohm-meter.
 $R_1 = \dots\dots\dots$ ohms.
- 2) Similarly determine the inductance of the coil by an L-C meter.
 $L_1 = \dots\dots\dots$ henry, $XL_1 = 2\pi fL_1 = \dots\dots\dots$ ohms.

- 3) Now take the remaining two coil terminals and measure its resistance and inductance.
 $R_2 = \dots\dots\dots \text{ohms}$, $L_2 = \dots\dots\dots \text{henry}$, $XL_2 = \dots\dots\dots \text{ohms}$.

Determine R/XL_1 and R/XL_2 . From this ratio determine the running winding and starting winding terminals.

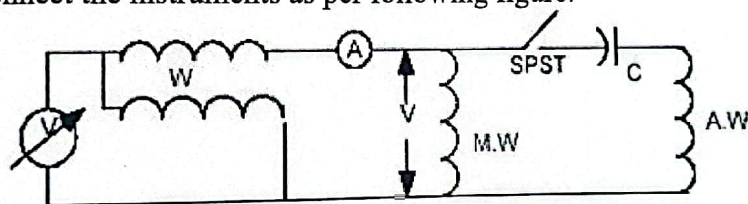
B) Running of the motor by the main winding:

- 1) Connect the terminals of the main winding as per following figure.



C) Running of the motor with the auxiliary winding:

- 1) Connect the instruments as per following figure.



Report:

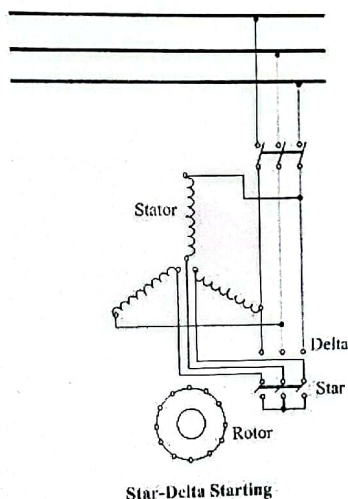
1. In step-B why did the motor rotate in both directions?
2. What is the purpose of capacitor in step-C ? Explain.
3. What have you noticed when you switched-off the SPST in step-C ?
4. Why does the motor continue to run in step-C, when the SPST was switched-off?

Experiment No: 06

Name of Experiment: Starting an Induction Motor by applying star-delta starter.

Theory: The induction motor must be built to run normally with a mesh-connected stator winding. At starting, the winding is connected temporarily in star. The phase voltage is thus reduced to $1/\sqrt{3} = 0.58$ of normal, and the motor behaves as if the auto-transformer were employed with a ratio $x = 0.58$. The starting current per phase is $I_s = 0.58I_x$, the line current is $(0.58)^2 I_x = 0.33I_x$, the starting torque is one-third of short-circuit value.

The method is cheap and effective, so long as the starting torque is not required to exceed about 50 percent of full load torque. It can therefore be used for machine-tools, pumps, motor generators etc. The method is unsuitable for motors at voltages exceeding 3000 V. because of the excessive number of stator turns needed for delta running. Where induction motors are required to run for considerable periods on small loads, a star-delta switch permits the machine to be star-connected during these periods, with reduction of magnetizing current and increase in efficiency.



This is a starting method that reduces the starting current and starting torque. At starting the Induction motor is star connected after it reached the approximate operational speed it is switch to delta. The motor must be delta connected during a normal run, in order to be able to use this starting method. This starting method only works when the application is light loaded during the start. If the motor is too heavily loaded, there will not be enough torque to accelerate the motor up to speed before switching over to the delta position.

Apparatus:

- 1) 3- ϕ Induction Motor
- 2) Star-Delta Starter
- 3) Connecting Wires

Table: (Readings): Same as previous experiments